Closing the center-embedding gap in Spanish: can grammatical features do the work of memory?

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Abstract

Center embeddings (clauses nested inside one another in the left branches of the structure) push comprehension to its limits, making them a natural probe for the memory architecture that supports language (**Karlsson2007**). Their signature is the missing-VP illusion: when one of three required verbs is deleted (see below for examples), readers accept the ungrammatical sentence more readily than the grammatical one, as if a stored prediction had silently decayed (**Lewis2005**; **Hahn2022**). The illusion was recently demonstrated in Spanish (**Paneda2024**). However, illusion magnitude tracked the statistical rarity of the verb sequence, while individual working-memory span predicted nothing. Storage capacity, on its own, appears to be the wrong variable.

Spanish exhibits relativizers that differ in morphological content while occupying identical syntactic positions: featureless *que* versus *el cual* and *quien*, which carry gender, number and animacy. If the illusion arises because degraded memory traces are reconstructed ambiguously, then feature-bearing relativizers should supply distinctive retrieval cues and reduce the illusion, with no change in sentence length, structure, or embedding depth. Featureless *que* should not.

We will test this in 120 speakers, combining speeded acceptability judgements, eye-tracking and EEG, and predict a graded reduction in illusion rate with feature richness, an attenuated sustained anterior negativity during maintenance, and a restored P600 at the missing verb.

The result would place grammatical features, not capacity, at the centre of how the brain holds structure open—and hands a broadly neurobiological question, what makes a memory cue distinctive, a linguistically precise manipulation.

Keywords: sentence processing; cue-based retrieval; working memory; morphosyntactic features; ERP; Spanish

SUPPORTING MATERIALS

The critical manipulation, on one item. All conditions are double center embeddings of identical depth; the featural content of the relativizer is crossed with grammaticality (VP2 entrevistó present vs. deleted).

(1) a. La política **a la que** el periodista **al que** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la que** el periodista **al que** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
BASELINE — *que*, gender/number only via the article; the condition used in all prior work.

(2) a. La política **a la cual** el periodista **al cual** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la cual** el periodista **al cual** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
FEATURE-BEARING — *el cual* inflects for gender and number on the pronoun itself.

(3) a. La política **a quien** el periodista **a quien** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a quien** el periodista **a quien** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
ANIMACY-BEARING — *quien* restricts the antecedent to a human referent.

Differential object marking on *a* blocks a subject-relative reading throughout, so no condition is ambiguous at the relativizer.

REFERENCES